

۲ ضربی سری

حالا با طانی $\sum_{n=-\infty}^{\infty} x[n] e^{j\omega n}$

1 Hilbert Transform

$$1. x_1(t) = \text{sinc}(\pi w t)$$

$$\hat{x}_1(t) = x_1(t) * \frac{1}{\pi t} = \mathcal{F}^{-1} \{ \hat{X}_1(f) \}$$

$$= \mathcal{F}^{-1} \left\{ \mathcal{F} \{ x_1(t) \} \mathcal{F} \left\{ \frac{1}{\pi t} \right\} \right\} = \mathcal{F}^{-1} \left\{ \frac{1}{\pi w} \text{rect} \left(\frac{f}{\pi w} \right) \times -j \text{sgn}(f) \right\}$$

$$I = \left[\text{rect} \left(\frac{f - \frac{w}{\pi}}{w} \right) - \text{rect} \left(\frac{f + \frac{w}{\pi}}{w} \right) \right] \frac{1}{\pi w} j$$

Shift Property (second property) $\Rightarrow$

$$\mathcal{F}^{-1} \{ I \} = \frac{1}{\pi j} (e^{j\pi t w} - e^{-j\pi t w}) \text{sinc}(w t)$$

$$\rightarrow x_1(t) = \mathcal{F}^{-1} \{ I \} = \text{sinc}(w t) \sin(\pi t w) = \frac{\sin(\pi w t)}{\pi w t} = \frac{1 - \cos(2\pi w t)}{2\pi w t}$$

$$2. x_r(t) = A \cos(\pi f_0 t + \theta) \quad \pi f_0 = w_0$$

$$\begin{aligned} \hat{X}_r(w) &= A \pi e^{-j\theta w} (\delta(w - w_0) + \delta(w + w_0)) (-j \text{sgn}(w)) \\ &= A e^{-j\theta w} \frac{\pi}{j} (\delta(w - w_0) - \delta(w + w_0)) \end{aligned}$$

$$\xrightarrow{\mathcal{F.T}^{-1}} x_r(t) = A \sin(w_0 t + \theta) = A \sin(\pi f_0 t + \theta)$$

$$3. x_3(t) = A \pi\left(\frac{t}{T}\right)$$

$$\hat{x}_3(t) = x_3(t) * \frac{1}{\pi t} = \frac{A}{\pi} \int_{-\infty}^{\infty} \pi\left(\frac{\tau}{T}\right) \frac{1}{t-\tau} d\tau$$

$$= \frac{A}{\pi} \int_{-T/2}^{T/2} \frac{1}{t-\tau} d\tau = \frac{A}{\pi} \ln \frac{t+T/2}{t-T/2} \quad \checkmark$$

$$(a) \quad v(t) = A = \lim_{T \rightarrow \infty} A \pi\left(\frac{t}{T}\right) = \lim_{T \rightarrow \infty} \hat{x}_3(t)$$

$$\rightarrow \hat{v}(t) = \lim_{T \rightarrow \infty} \hat{x}_3 = \frac{A}{\pi} \lim_{T \rightarrow \infty} \ln 1 = 0 \quad \checkmark$$

$$4. x_\xi(t) = \frac{1}{(t-a)^n}$$

$$\frac{d}{dt} \left(\frac{1}{t^{n+1}} \right) = -(n+1) \frac{1}{t^{n+2}} \rightarrow F \left\{ \frac{d}{dt} \frac{1}{t^{n+1}} \right\} = -(n+1) F \left\{ \frac{1}{t^{n+2}} \right\}$$

$$F \left\{ \frac{d}{dt} t \right\} = j\omega F(\omega) \Rightarrow F \left\{ \frac{1}{t^{n+1}} \right\} = - \frac{j\omega}{n+1} F \left\{ \frac{1}{t^n} \right\}$$

$$F \left\{ \frac{1}{t} \right\} = -j\pi \operatorname{sgn}(\omega) \rightarrow F \left\{ \frac{1}{t^n} \right\} = \frac{-j\pi (-j\omega)^{n-1}}{(n-1)!} \operatorname{sgn}(\omega)$$

$$\text{shift property, } F \left\{ \frac{1}{(t-a)^n} \right\} = \underbrace{\frac{-j\pi (-j\omega)^{n-1}}{(n-1)!} \operatorname{sgn}(\omega)}_I e^{-j\omega a}$$

I

$$\rightarrow F \left\{ \hat{x}_\xi(t) \right\} = -j\pi \operatorname{sgn}(\omega) I = \frac{-\pi}{(n-1)!} (-j\omega)^{n-1} e^{-j\omega a}$$

$$\rightarrow \hat{x}_\xi(t) = \frac{-\pi}{(n-1)!} F \left\{ (-j\omega)^{n-1} e^{-j\omega a} \right\}$$

$$\rightarrow \hat{x}_5(t) = -\frac{r}{(n-1)!} \delta(t-a) (-1)^n$$

$$5. x_5(t) = \cos^4(\omega t)$$

$$\cos^4(\omega t) = \left(\frac{1 + \cos 2\omega t}{2} \right)^2 = \frac{1}{4} + \frac{\cos 2\omega t}{2} + \frac{\cos^2 2\omega t}{4}$$

$$= \frac{3}{8} + \frac{1}{2} \cos 2\omega t + \frac{1}{8} \cos 4\omega t$$

$$\text{part 2} \rightarrow H\{\cos At\} \rightarrow \sin At, H\{A\} = 0$$

$$\Rightarrow H\{x_5(t)\} = \frac{1}{4} \sin 2\omega t + \frac{1}{8} \sin 4\omega t$$

$$6. x_6(t) = \frac{t}{(a^2+t^2)^r}$$

$$H\left\{\frac{d}{dt} x\right\} = \frac{d}{dt} H\{x(t)\}, \frac{t}{(a^2+t^2)^r} = -\frac{1}{r} \frac{d}{dt} \frac{1}{a^2+t^2}$$

$$\rightarrow \hat{x}_6(t) = -\frac{1}{r} \frac{d}{dt} H\left\{\frac{1}{a^2+t^2}\right\} = -\frac{1}{r} \frac{d}{dt} \frac{1}{a(a^2+t^2)}$$

$$= \frac{t - a^2}{2a(a^2+t^2)^2}$$

2 Hilbert Transform Properties

1. $x(t)$ is even $\rightarrow x(t) = x(-t)$

$$\rightarrow X(f) = \int_{-\infty}^{\infty} x(t) e^{-j\pi f t} dt = \int_{-\infty}^{\infty} x(-t) e^{-j\pi f t} dt$$

$$t = -u \rightarrow dt = -du \rightarrow X(f) = - \int_{\infty}^{-\infty} x(u) e^{-j\pi f (-u)} du$$

$$= \int_{-\infty}^{\infty} x(u) e^{j\pi f u} du = X(-f) \rightarrow X(f) \text{ is also even}$$

Same for odd $\Rightarrow$ if $x(t) = -x(-t) \Leftrightarrow X(f) = -X(-f)$
 approach

$$F\{H\{x_{\text{even}}\}\} = \underbrace{F\{x_{\text{even}}\}}_{\text{even}} \times \underbrace{F\left\{\frac{1}{\pi t}\right\}}_{\text{odd}} = \text{odd}$$

$$\rightarrow H\{x_{\text{even}}\} = \text{odd} \checkmark$$

$$F\{H\{x_{\text{odd}}\}\} = \underbrace{F\{x_{\text{odd}}\}}_{\text{odd}} \times \underbrace{F\left\{\frac{1}{\pi t}\right\}}_{\text{odd}} = \text{even}$$

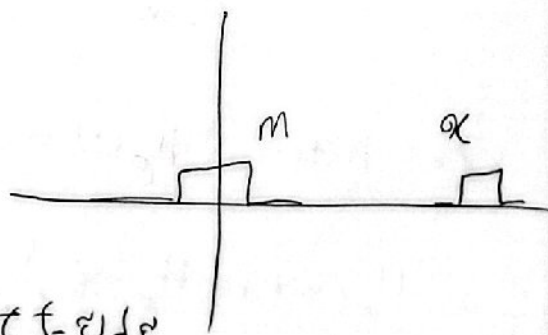
$$\rightarrow H\{x_{\text{odd}}\} = \text{even} \checkmark$$

2.

$$F\{c(t)\} = F\{m\} * F\{x\}$$

$$= \int_{-\infty}^{\infty} M(\tau) X(f-\tau) d\tau$$

$$F\{\hat{c}(t)\} = -j\pi \operatorname{sgn} f \int_{-\infty}^{\infty} m(\tau) X(f-\tau) d\tau$$



Non overlapping $\rightarrow$

$$F\{\hat{c}\} = \int_{-\infty}^{\infty} m(\tau) \cdot j\pi \operatorname{sgn}(t-\tau) X(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} m(\tau) \hat{X}(t-\tau) d\tau = M * \hat{X}$$

$$\rightarrow \hat{c} = m(t) \hat{x}(t)$$

3 Even-odd decomposition

1.

causality $\rightarrow$

$$x(t) = h_e(t) u(t) \begin{cases} t < 0 & u=0 \rightarrow I=0 \\ t > 0 & u=1, h_e = \frac{h(t) + h(-t)}{2} = \frac{h(t)}{2} \\ & \rightarrow I = h(t) \end{cases}$$

$$\rightarrow I = \begin{cases} 0 & t < 0 \\ h(t) & t > 0 \end{cases} \quad h(t) = 0 \text{ for } t < 0 \rightarrow I = \begin{cases} h(t) & t > 0 \\ h(t) & t < 0 \end{cases}$$

$$\rightarrow I = x(t) = h_e(t) u(t) = h(t)$$

$$2. \quad h_e = \int_{-\infty}^{\infty} H_e(f) e^{j2\pi ft} df$$

$$3. \quad h(t) = x(t) = h_e(t) u(t) \xrightarrow{F.T} H(f) = \frac{1}{2} H_e * \left(\frac{1}{j\omega} + \pi \delta(\omega) \right)$$

$$= H_e(\omega) + H_e * \frac{1}{j\omega}$$

$$4. \quad h_0 = h - h_e = \gamma h_e u(t) - h_e = h_e (\gamma u(t) - 1) \\ = h_e \operatorname{sgn}(t)$$

$$5. \quad h_0 = h_e \operatorname{sgn}(t) \xrightarrow{f.T} H_0 = H_e(f) * f\{\operatorname{sgn}(t)\} \\ = H_e * \frac{1}{j\omega} = H_e * \frac{1}{j\pi f}$$

$$6. \quad H_0 = H_e * \frac{1}{j\pi f} = \frac{1}{j} \underbrace{(H_e * \frac{1}{\pi f})}_{\hat{H}_e} = -j \hat{H}_e(f)$$

4. wireless connection

$$a) \quad L_{fs} = 40 \log d + 20 \log f - 147.55$$

$$\text{user}_1: L_{sf_1} = 20 \log(10^3) + 20 \log(2.1 \times 10^9) - 147.55$$

$$= 20 (\log 10^3 + \log 10^9 + \log 2.1) - 147.55$$

$$= 20 \times 246.44 - 147.55 \approx 98.8943$$

$$\rightarrow P_{\text{received}_1} = 20 - 98.89 = -78.89 \text{ dBm} \rightarrow \text{a connection}$$

can be established. ✓

user 2:

$$L_{fs2} = 20 (4 + 9 + \log(2.1) + \log 2) - 147.55 = 104.91$$

$$P_{received2} \approx 36 - 104.91 \approx -68.91 > -70$$

→ a connection can be established.

$$b) P_{new1} = P_{min} = -70 \text{ dB}$$

$$\rightarrow a_{dB} = -70 - 78.89 = 8.89 \text{ dB}$$

$$\rightarrow a = 10^{a_{dB}/10} \approx 7.74 \text{ (as given)}$$

$$c) P_{loss, new} = 20 + 70 = 90 \text{ dB}$$

$$= 20(\log(f) + 3) - 147.55$$

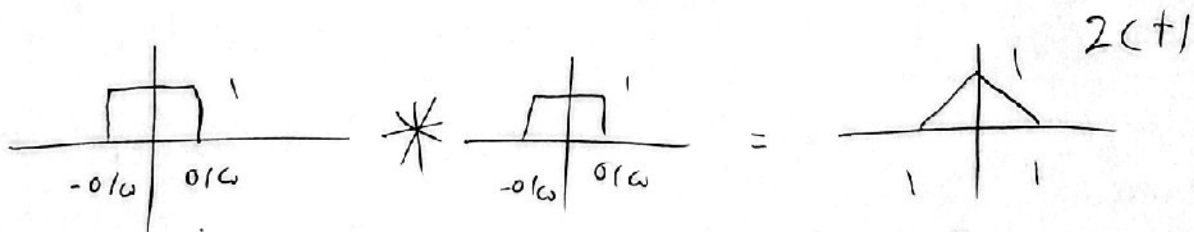
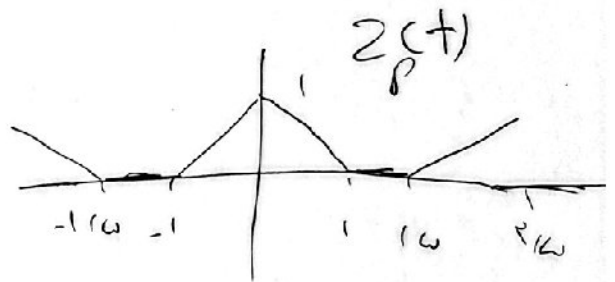
$$\rightarrow \log f = 8.87 \rightarrow f \approx 754.22 \text{ MHz}$$

5 Attenuation and Amplification

$$1. Z_p(t) = x(t) + o/w$$

$$\text{F.T} \rightarrow Z_p(f) = X(f) + o/w \delta(f)$$

$$\rightarrow X(f) = Z_p(f) - o/w \delta(f)$$



$$\rightarrow F\{Z(t)\} = F\{\text{rect} * \text{rect}\} = F\{\text{rect}\} \times F\{\text{rect}\}$$

$$= \text{sinc}(f)$$

$$z_p(t) = \sum_{k \in \mathbb{Z}} z(t + 2.5k) = z(t) * \sum_{k \in \mathbb{Z}} \delta(t - 2.5k)$$

$$\rightarrow F\{z_p(t)\} = z(f) \times \frac{1}{T} \sum_{k \in \mathbb{Z}} \delta(f - \frac{1}{T}k)$$

$$= \frac{1}{3} \sum_{k \in \mathbb{Z}} z(f) \Big|_{f = \frac{k}{3}} = \frac{1}{3} \sum_{k \in \mathbb{Z}} z\left(\frac{k}{3}\right)$$

$$= \frac{1}{3} \sum_{k \in \mathbb{Z}} \text{sinc}^2\left(\frac{k}{3}\right) \delta\left(f - \frac{k}{3}\right)$$

$$\rightarrow x(f) = \frac{1}{3} \left(\sum_{k \in \mathbb{Z}} \text{sinc}^2\left(\frac{k}{3}\right) \delta\left(f - \frac{k}{3}\right) \right) = \delta(f) \frac{1}{3}$$

$$2. P_{\text{av}} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T |x(t)|^2 dt = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt$$

$\uparrow$
 x is periodic

$$T_0 = 3 \rightarrow P_{\text{av}} = \frac{1}{3} 2 \left(\int_{-1/\omega}^1 (-0.1\omega)^2 dt + \int_{-1}^0 (t + 0.1\omega)^2 dt \right)$$

$$= \frac{1}{3} 2 \left(\frac{1}{8} + \frac{1}{12} \right) = \frac{5}{36} \checkmark$$

$$3. P_{\text{av}} = \frac{5}{36} \text{ W} \rightarrow P_{\text{av}} = 10 \log \frac{5}{36} = -8.57$$

$$P_{\text{in, section 1}} = 20 \text{ uW} \quad \text{~~not log~~}$$

$$\rightarrow P_{\text{in, min}} = 10 \log 2 \times 10^{-5} = -46.98$$

$$\rightarrow \text{setting } P_{\text{av}} - 4 \text{ dB} \times L = P_{\text{in, min}}$$

$$\rightarrow L = \frac{38.4}{4} = 9.6 \text{ km}$$

$$\rightarrow \text{minimum number of sections} \rightarrow \frac{500}{9.6} = 52 \quad \boxed{26}$$

$$B_{\text{dB}} = 38.4 \text{ dB} \checkmark$$

$$\rightarrow B = 10^{\frac{38.4}{10}} = 6948.64 \quad \boxed{}$$